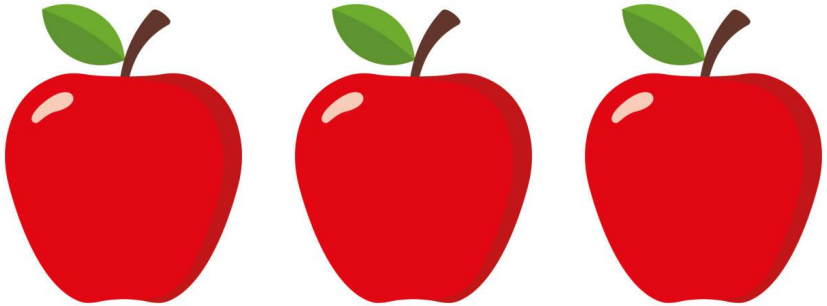
From 0 to $\mathbb{C}$

$$0 \longrightarrow \mathbb{N} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R} \longrightarrow \mathbb{C}$$

Chae Young Han
(SKKU)
2025.12.01.
MIMIC

$$\boxed{3}i + 1 \qquad \pi = \boxed{3.}1415926\cdots \qquad \frac{2}{\boxed{3}}$$



There are 3 apples.

3=2+1

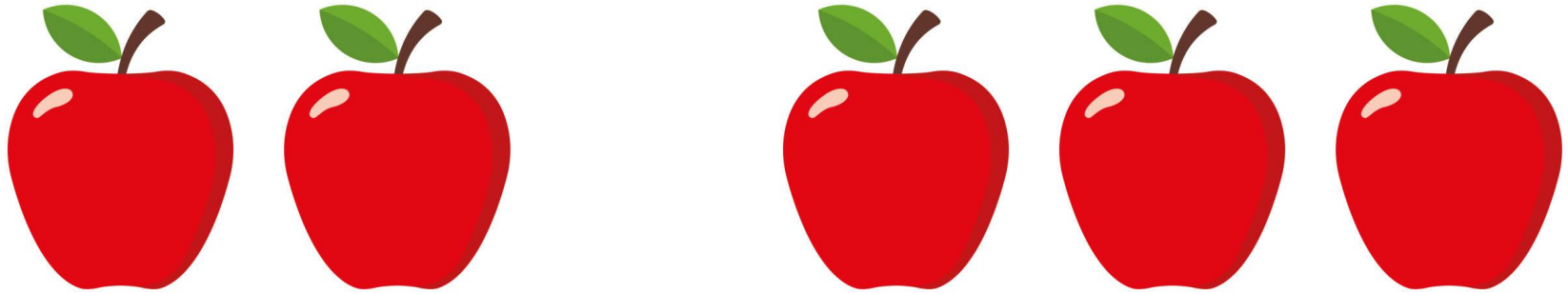
2=1+1

1=1+0

What is 0?

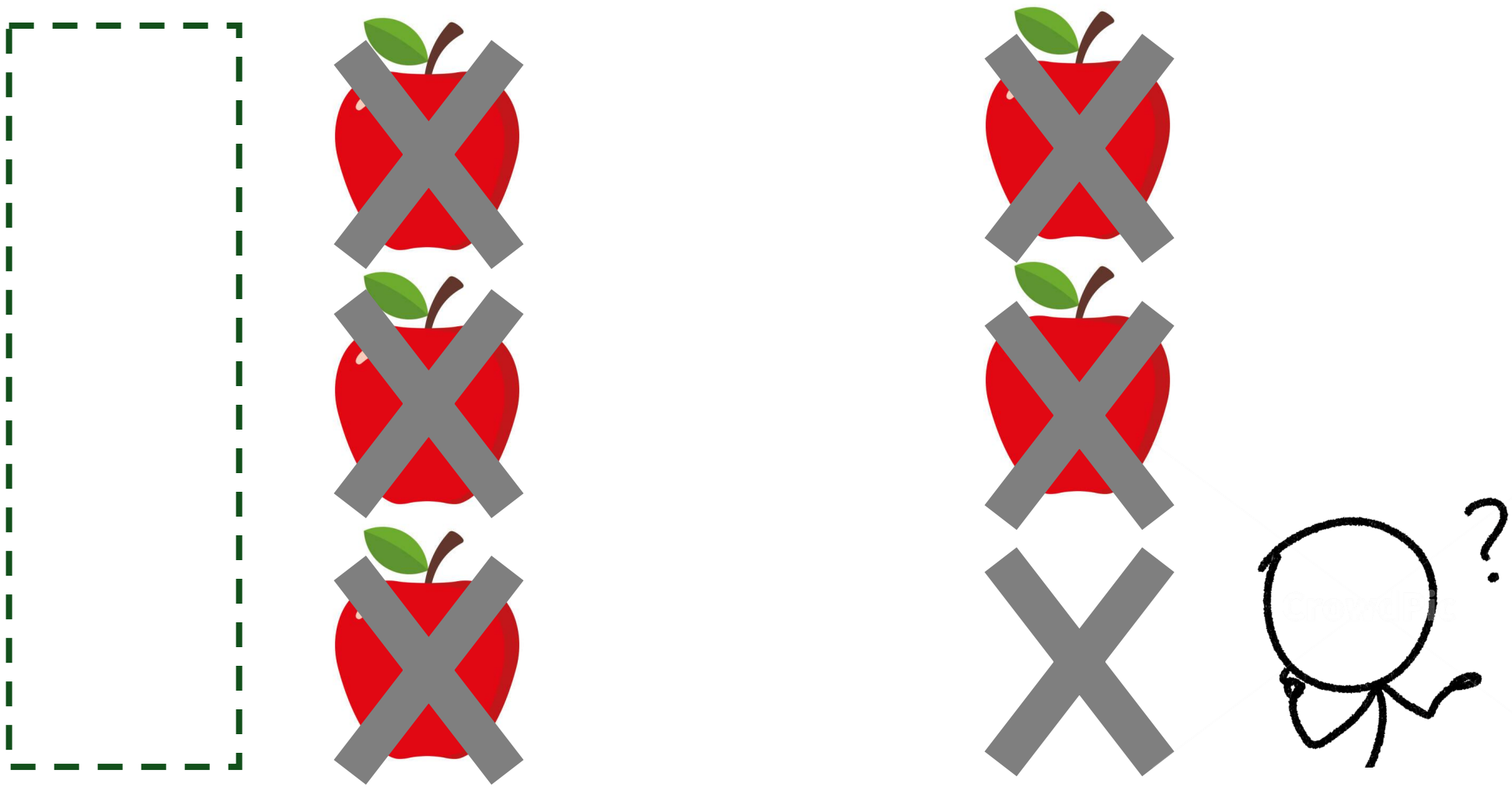
Assume that all the numbers are disappeared.

$$\Rightarrow \quad \mathbb{N} = \{0, 1, 2, 3, \cdots\}$$



2 + 3

$$x + 3 = 2$$



How can we construct $\mathbb{Z}$?

$$\langle a, b\rangle \sim \langle c, d\rangle \;\Leftrightarrow\; a + d = b + c$$

$$\mathbb{Z} = (\mathbb{N} \times \mathbb{N})/\sim$$

e.g
.

$$3 + 0 = 1 + 2 \;\Leftrightarrow\; \langle 3, 1\rangle \sim \langle 2, 0\rangle \quad \Rightarrow \quad \langle 3, 1\rangle \in [\langle 2, 0\rangle]$$

$$0 + 3 = 2 + 1 \;\Leftrightarrow\; \langle 0, 2\rangle \sim \langle 1, 3\rangle \qquad\qquad \langle 1, 3\rangle \in [\langle 0, 2\rangle]$$

How can we construct $\mathbb{Z}$?

$$\langle a, b\rangle \sim \langle c, d\rangle \;\Leftrightarrow\; a + d = b + c$$

$$\mathbb{Z} = (\mathbb{N} \times \mathbb{N})/\sim$$

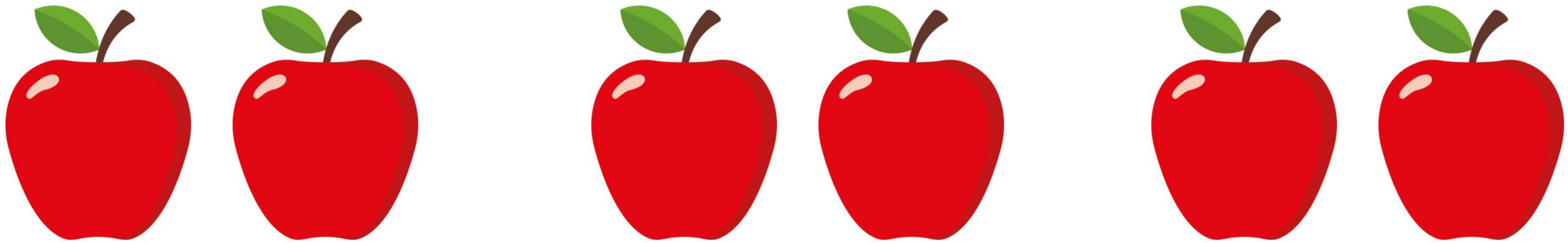
$$\boxed{\begin{aligned} n &= [\langle n, 0\rangle] \\ -n &= [\langle 0, n\rangle] \end{aligned}}$$

e.g
.

$$3 + 0 = 1 + 2 \;\Leftrightarrow\; \langle 3, 1\rangle \sim \langle 2, 0\rangle$$
$$0 + 3 = 2 + 1 \;\Leftrightarrow\; \langle 0, 2\rangle \sim \langle 1, 3\rangle$$

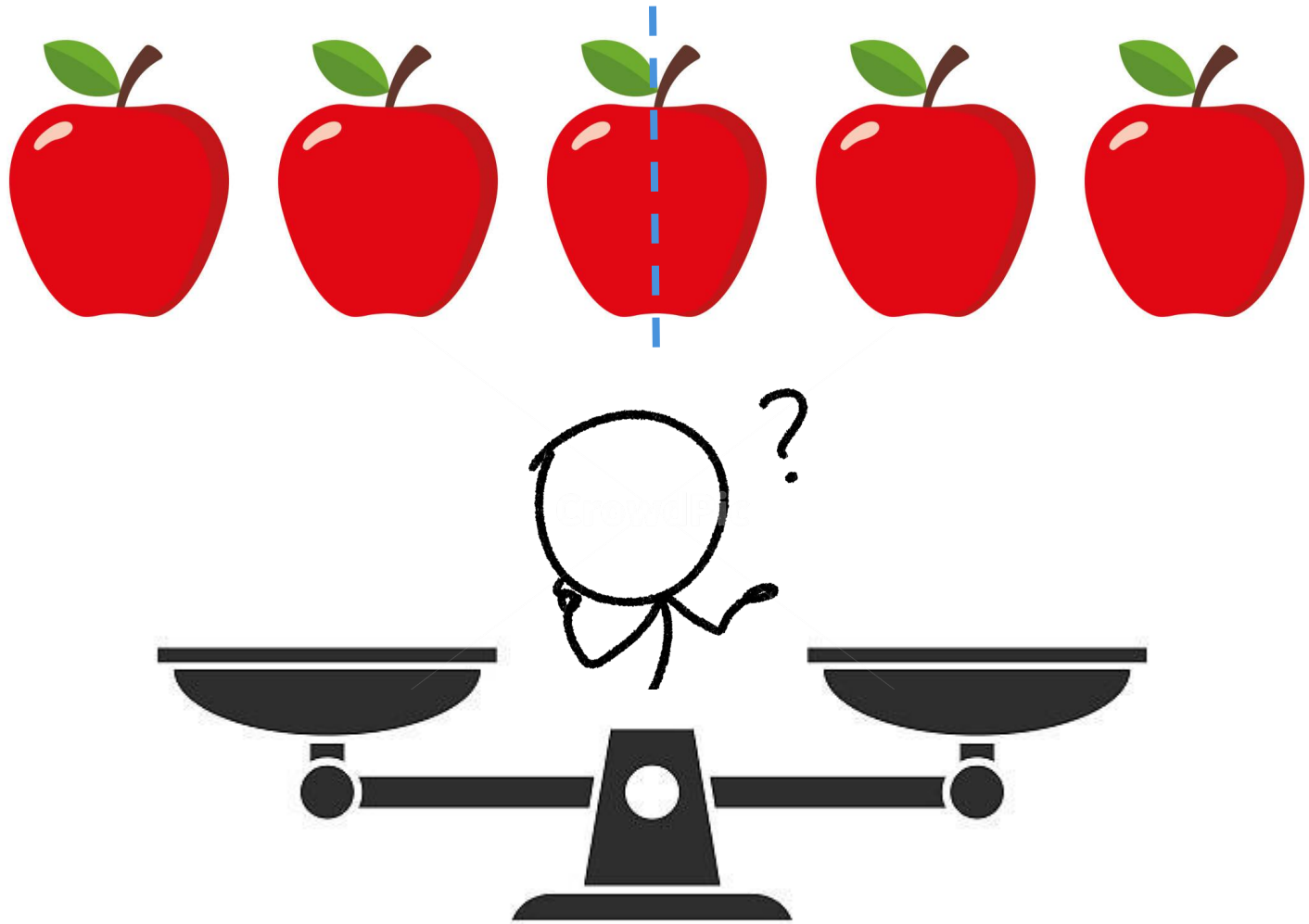
$\Rightarrow$

$$\langle 3, 1\rangle \in [\langle 2, 0\rangle] = 2$$
$$\langle 1, 3\rangle \in [\langle 0, 2\rangle] = -2$$



2 X 3

$$x \times 2 = 5$$



How can we construct $\mathbb{Q}$?

$$\langle a, b\rangle \sim \langle c, d\rangle \;\Leftrightarrow\; ac = bd$$

$$\mathbb{Q} = (\mathbb{Z} \times \mathbb{Z} \setminus \{0\})/\sim$$

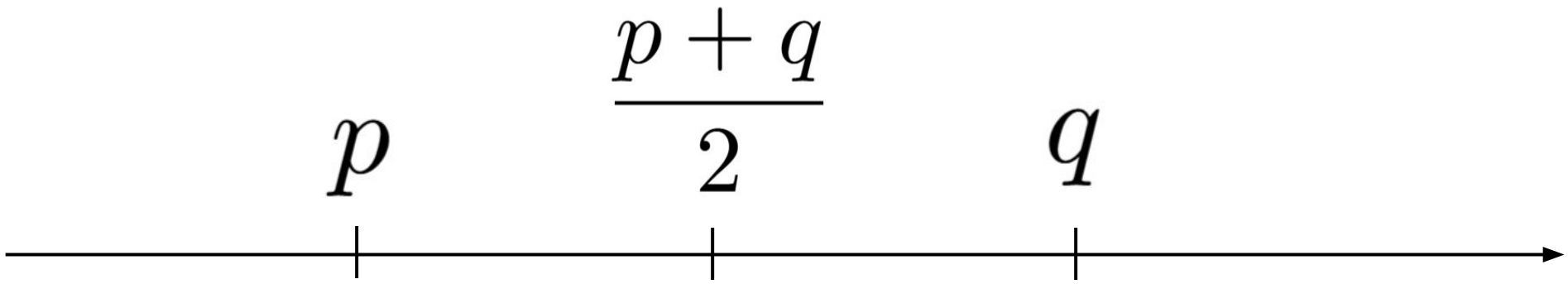
$$\boxed{\frac{p}{q} = [\langle p, q\rangle]}$$

e.g

.

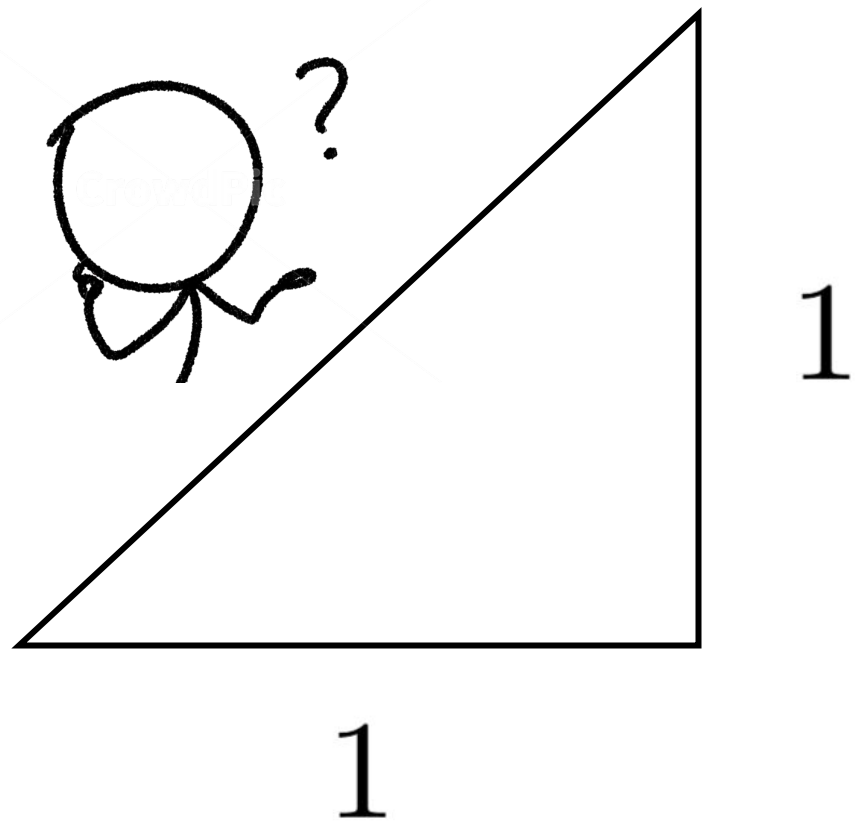
$$3 \cdot 10 = 6 \cdot 5 \;\Leftrightarrow\; \langle 3, 5\rangle \sim \langle 6, 10\rangle \;\Rightarrow\; \langle 6, 10\rangle \in [\langle 3, 5\rangle] = \frac{3}{5}$$

$$\langle 3, 5\rangle \in [\langle 6, 10\rangle] = \frac{6}{10}$$



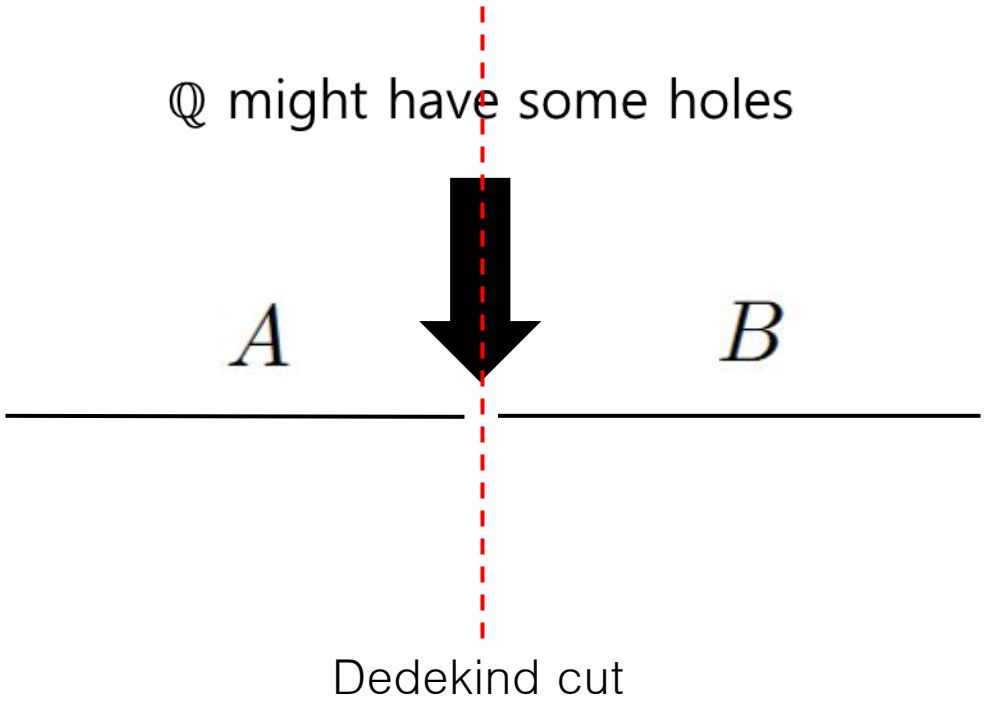
It seems like $\mathbb{Q}$ is dense enough, but…

$$x^2 = 2$$



How can we construct $\mathbb{R}$?

i) Dedekind cut



1. $A \neq \varnothing$
   $A \neq \mathbb{Q}$
2. For $x, y \in \mathbb{Q}$ with $x < y$, $y \in A \to x \in A$.
   If $x \in A$, then there exists $y \in A$ such that $y > x$.
3. e.g
4. $\sqrt{2} = \{q \in \mathbb{Q} \mid q^2 < 2 \text{ or } q < 0\}$

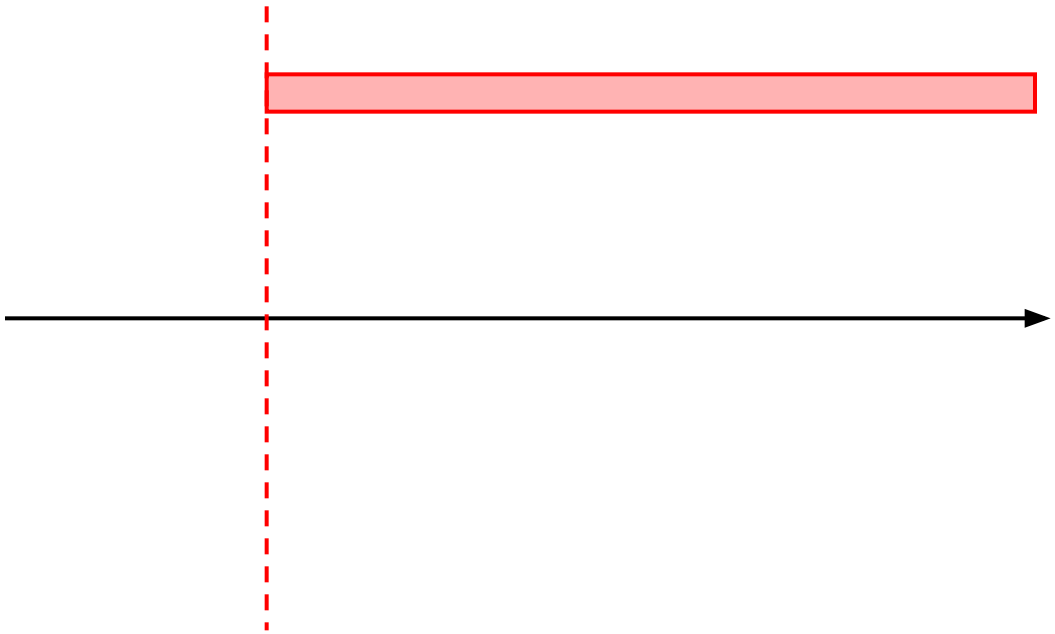
$$2 = \{q \in \mathbb{Q} \mid q < 2\}$$

How can we construct $\mathbb{R}$?

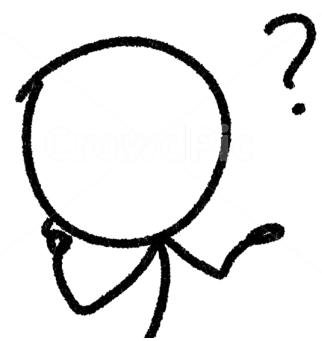
ii) Cauchy sequence

$$1$$
$$1.4$$
$$1.41$$
$$1.414$$
$$1.4142$$
$$\vdots$$

$$\forall \varepsilon > 0,\ \exists N,\ \forall m, n \geq N,\ |a_m - a_n| < \varepsilon$$

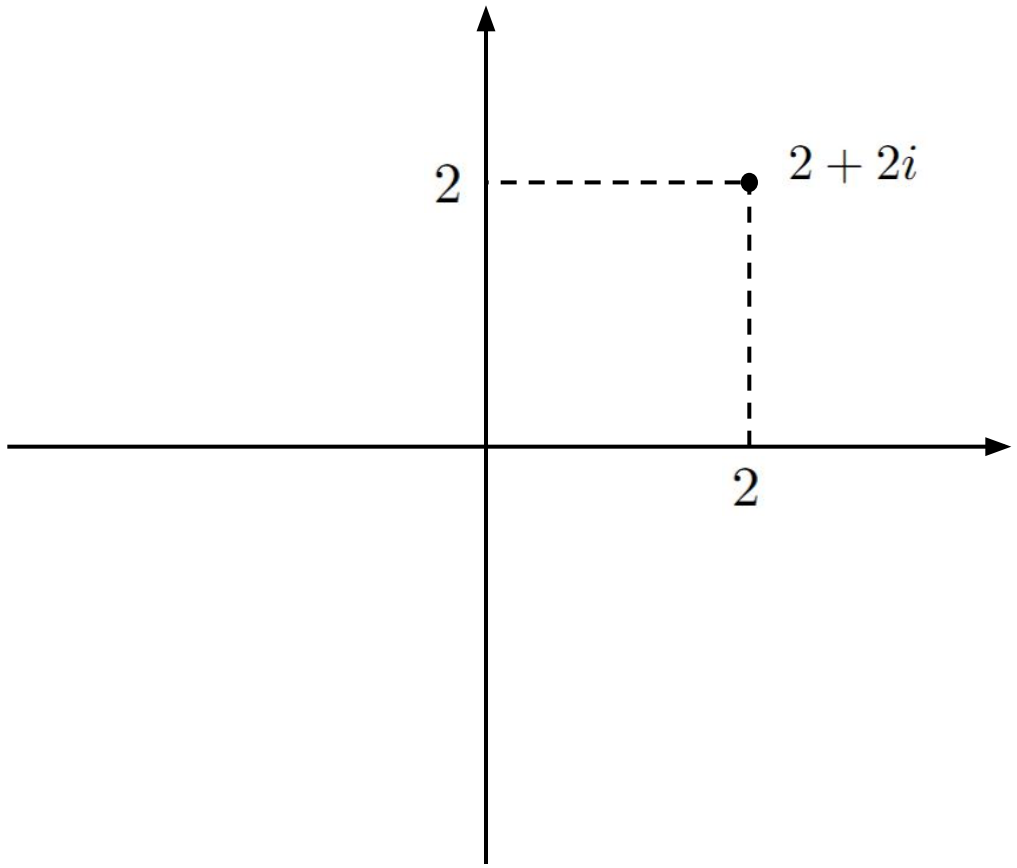


$$x^2 + 1 = 0$$



How can we construct $\mathbb{C}$?

$$\mathbb{C} = \{a + bi \mid a \in \mathbb{R}, b \in \mathbb{R}\}$$



Beyond the Complex numbers

Quaternion

$$\{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\}$$

$$i^2 = j^2 = k^2 = ijk = -1$$

$$ij = k, \quad jk = i, \quad ki = j$$

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Thank you